



Péter Juhász

PhD Researcher in Mathematics

CONTACT

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EDUCATION

2022 – 2025: PhD in Mathematics
Aarhus University, Denmark

2019 – 2021: MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016: MSc in Physics
Budapest University of Technology and Economics, Hungary

2013 – 2016: BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

Stochastic Processes
Probability Data Analysis
Python C++ SQL

LANGUAGES

Hungarian

English

German

CERTIFICATIONS

2021: Software Architect ISAQB

2021: Software Designer Bosch

PROFILE

I am a PhD Researcher in Mathematics at Aarhus University, graduating in December 2025. My research consists of stochastic modeling of high-dimensional random systems supported by large-scale C++/Python simulations. Previously, I worked at Bosch as a deep learning researcher and software architect in automated driving, focusing on perception, prediction, decision-making, and scalable system design.

PROFESSIONAL EXPERIENCE

PhD Researcher
Aarhus University, Denmark

2022 – 2025

- Developed and analyzed **stochastic models for complex, high-dimensional systems**, with applications to interacting stochastic systems.
- Designed **efficient C++ and Python algorithms** for large-scale stochastic network models.
- Proved **theoretical results in stochastic point processes and random graphs**, contributing to the mathematical foundations of probabilistic modeling.
- Taught **advanced probability, data analysis, and differential equations** to mathematics and data-science students.

Deep Learning Researcher & Software Architect
Bosch Group, Hungary

2018 – 2022

- Patented a **stochastic trajectory prediction model** improving accuracy by 3× and uncertainty estimation by 10× over baseline.
- Led a **team of 11** developing environment prediction modules for automated driving as a technical lead.
- Created a **framework for automated ground-truth generation** enabling large-scale data-driven model evaluation.

Software Engineer
Ericsson Telecommunications, Hungary

2016 – 2018

- Designed **response time-based load-balancing algorithms** for distributed systems, improving efficiency through quantitative optimization.

AWARDS, GRANTS

- 2022: PhD Fellowship

Danish Data Science Academy

1 890 000 DKK
- 2021: Excellence Award

Hungarian National Bank

350 000 HUF
- 2025: Individual Travel Grant

European Mathematical Society

500 EUR
- 2024: Academic Mobility Grant

Aarhus University

20 000 DKK



SELECTED PUBLICATIONS, PATENTS

- C. Hirsch, B. Jahnel, and P. Juhász.
Functional limit theorems for edge counts in dynamic random connection hypergraphs,
arXiv preprint (2025), arxiv:2507.16270
- P. Juhász.
Method and device for predicting the trajectory of a traffic participant, and sensor system,
CNIPA: CN113988353A, EPO: EP3916697A1
JPO: JP2021190119A USPTO: US2021366274A1 (2021)
- P. Juhász.
Information propagation in stochastic networks,
Phys. A (577) (2021), 126070
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay.
A Bayesian approach to identify Bitcoin users,
PLoS One (13) (2018), e0207000

TEACHING

Advanced Probability Theory 10 ECTS Exercise Class	2025
Data Project 10 ECTS Computer Project Class	2024
Ordinary Differential Equations 5 ECTS Exercise Class	2023